

Table 1: Overall contact fidelity and dynamic quality on CORE4D-G1 (52 sequences / 11 objects). $\uparrow$: higher is better, $\downarrow$: lower is better. **Bold** = best, underline = second best. Ours = SPIDER-CEM. Contact metrics are penetration-aware (a loose "any-contact" count that rewards interpenetration is intentionally omitted). Dynamic-quality metrics for Ours, SBTO and OmniRetarget are computed on a MuJoCo physics rollout (native for the two optimizers; position-servo replay for the kinematic OmniRetarget, E197 protocol) via the same code, hence directly comparable; GMR's are from its own kinematic pipeline at a different scale and are indicative only (read jointly with contact).

Metric	Ours	SBTO	OmniRt	GMR
Phys. contact@3mm $\uparrow$	0.588	<u>0.172</u>	0.101	0.001
Phys. pen@3mm $\downarrow$	<u>0.176</u>	0.479	0.536	0.046
Geom. pen@2mm $\downarrow$	0.052	0.427	0.381	<u>0.059</u>
Ankle jerk p95 (m/s^3) $\downarrow$	<u>653.8</u>	1059.4	1560.8	170.4
Obj speed max (m/s) $\downarrow$	1.29	<u>1.83</u>	2.83	1.88
Foot slip max (m) $\downarrow$	0.87	1.16	1.33	<u>0.96</u>